

Hi everyone and
welcome to week 9
of the Human-AI
Interaction course.

Today we're going to be
talking about another
dimension of human-centered
values in the design
of AI systems and
specifically we're going to
be talking about
environmental sustainability.

Okay, so this video is going
to be focused mostly on
generative AI. So a lot of
the stuff when I mention
AI in general, I mean
generative AI. and we're
especially going to look at
this because as you might know
generative ai has a lot
of infrastructure behind
it in order to to be able
to function the way we would
like to so and as the scale
of that has been growing
in complexity then that's
the scale and the complexity
have been growing then
the resource and the
impact on the environment
have also been becoming quite
significant and this impact
in terms is in terms of
energy consumption water
and minerals and then And
accordingly, this kind of
leads to carbon emissions.

So, of course,
just to be clear,
giving you this lecture,
I'm not trying to say that
generative AI is not useful
or it's not kind of has
a lot of potential to
help drive sustainable
development goals. Of course
it does. But some people,
and I think it's an important
discussion to be had,

argue that the heavy resources of energy, use of resources like energy, water and minerals, spurred a movement that is called basically green AI. And this is a need basically to encompass, to estimate, analyze and eventually mitigate generative AI and environmental impact. So this is what we're going to be discussing today. I'm going to be focusing on how generative AI can support environmental sustainability, but we're mostly going to be looking at the discussions that are being had regarding green AI, basically understanding how to estimate, analyze, and then mitigate the effects that generative AI has on the environment. just to give you some context some figures to contextualize a little bit more the the discussion the background here you see i think it's one of google servers but what has been found from the international energy agency though in 2024 it estimated that data center's total electricity consumption will reach more than 1000 terawatts hours by 2026 which is more than double from what it was in 2022 and this growth directly relates to the need for larger data centers running high performance computers such as gpus an even newer version of this report so from 2025 mentions that the global total electricity consumption will reach more than 3500 kilowatt hours for the next three years so

much more than it was calculated before and this correspond almost to adding the equivalent of a japan in terms of energy consumption to the world's electricity consumption each year this this doesn't relate only to data centers but the data centers still play an important role in kind of contributing to this figure another kind of daunting figure that is similar like by 2028 the u.s data centers could end up be using up to 12 percent of the total u.s electricity consumption which is three times higher than what is now which is around 4.4 percent so it's not something we should be avoiding a discussion we should be avoiding basically and such substantial resource demands have delayed the retirement of things like coal plants and have intensified kind of conflicts with local communities that live around data centers and have their water or their energy grids kind of been balanced in places such as arizona and in spain and in ireland actually okay perhaps all of that you have heard before um there it's been quite a lot on the news for more closer to home perhaps in west london data centers seem to have imbalanced the grid so that new buildings cannot be raised and powered until the energy capacity is again rebalanced And it's not just energy. As I said, it's

also water for cooling
down these servers,
and which is especially
challenging as well.

And then for places
that are built with
scarcity of water
resources, that is actually
especially important,
right, to do that.

So what I want to focus
on today for the duration
of this class, for this
video, is on four parts. So
first we're going to give
some background information
about how do these numbers
come about, about how
do you end up calculating
AI energy use, and we're
going to bring some
examples for that. And then
we're going to look into
three research projects that
relate to how human-computer
interaction or human

-AI interaction can
contribute to this discussion
on sustainability for
generative AI, for green AI.

okay so generative
environmental
impact is actually
quite complex

not not only because we're
not only talking about
something monolithic and it's
actually quite a widespread
kind of life cycle process
so there are different
things for considerations
for calculating that energy
usage like training fine
-tuning inference these are
different kind of stages
that have different kind of
effects where it's different
if we're training or using
these models locally or on
there they're located in

remote servers on the cloud
it's different depending
on the time of day and the
kind of use and the region
that we're training or using
these models from the model
characteristics and the
architecture are also kind
of a big impact on this and
especially for inference
the task type that is being
kind of used for then that
also makes a huge difference
in how sustainable
environmentally sustainable
basically how much energy is
being used or not so some
pioneering papers in the
field conducted a life cycle
analysis on influential ai
models general ios the
first ones that was that
existed was mostly on bloom
176 billion parameter language
model and they found
that its training emitted
approximately 24.7 tons of
co2 equivalent gases and if we
consider only the dynamic
power consumption and around
50 tons if we account for
all the processes ranging
from equipment manufacturing
all the way to the base
operational consumption
so as you can see here in
their lifecycle analysis they
start from things like raw
material extraction and
materials manufacturing for
the servers for instance
these are not taking into
account but the equipment
manufacturing the model training
the model deployment
deployment and in this case
they have excluded again
the disposal or end-of-life
especially for the the
hardware aspect of things but

these three parts that are set in green end up consuming around 50 tons of CO₂ equivalent carbon emissions now there's not just about where what kind of hardware you're using there's a big difference if you're using cloud or local training so i don't know if you can see here this graph where you have the different countries and depending on their energy grid mix so basically does that location or does that country mostly have renewable energy or use like fossil fuels for for powering their country then or the region then it makes a huge difference in terms of how much CO₂ grams are emitted in this case it's about birth and you can see countries like Germany which have quite a high renewables usage in their energy mix then it's actually quite, sorry, Germany is here yeah, depending on the time of month then it actually makes a difference so France here which is very much nuclear powered is quite low in terms of CO₂ grams emitted But US and Germany and Australia, for instance, are very much more fossil fuel powered So that means that if we train in the cloud, rather than training locally on our own computers When if you would be locally, you would be only caring about the energy mix, for instance, of the UK Or England, but if you train in the cloud, you will have this flexibility of choosing which location to training

or to use basically for
your generative AI models
to have like this kind
of optimal grid mix and
also this also gives
us the opportunity of
batching resources like
batching tasks together
schedules together so
it's not just the country
though it's also like
the the time of use
so for instance in the
uk we have a lot of um
solar but also during
the summer months so
there's a lot of solar
but also there's a lot of
wind capacity so as you
can imagine that changes
depending on the day or
the season so maps like
these electricity maps
here actually give this
kind of prediction of
how much and how intense
is the carbon makes for
that specific country
and how much of it is
low carbon, how much of
it is renewable for a
specific timestamp. So
in theory, we can
develop systems that are
mostly focused and driven
towards renewables.
something we talked about,
I'm mentioning training a
lot, but I also mentioned
the inference. So generative
tasks are generally much
more energy and carbon
intensive depending to
discriminative tasks. So figure one
shows this kind of different
types of tasks like
text classification or token
classification or image
classification are significantly,
and this is logarithmic

scale, by the way. So there's significantly less emissions, grams of CO₂ equivalent in comparison to things like summarization or image generation, which is significantly more. And this is an experiment that Luciani and her colleagues did, where they actually are testing the average quantity of carbon emissions they produce for a thousand queries, if I'm not mistaken.

So there's a huge difference about what you're using your models for.

This is a little bit of background information about how generally these kinds of measures of sustainability are calculated. But I wanted to focus on this part two of this talk about how that relates to human AI interaction and how we can do something about that potentially in terms of as researchers that understand both the technology, but also the human behavior.

What I've proposed a lot of my research is looking at ecofeedback.

So we're going to look at what is ecofeedback and how does it relate to generative AI.

So most of you or many of you might have seen something like this. It's an in-house display for smart meters that depicts live data on your gas and electricity consumption.

And it's one of the most widely available examples of an ecofeedback device.

so this is what we call
a device that gives
feedback basically a little
visualization as you can see
here to encourage behavior
change and the hypothesis
with echo feedback is
that with more information
people will make more
sustainable choices in the
future and this is something
that has been studied
quite a lot in terms of
households as you can
imagine trying to make people
change when they're using
their electricity
when they're putting the
laundries when they're
putting their washing their
dishwashers and things
like this depending on how
much the grid is busy or
how is more convenient
in terms of pricing
so what we did in some recent
research was looking at
what kind of echo feedback
devices exist equivalent
for generative and general
actually ml tools so this is
not just generative ai tools
this is general and you
can see that there are quite
a few this was done in 2024
so we met we found around
16 tools but there have
been many more since as i
will add i will add a few more
in the next slides so
basically what these tools do
is that they they give this
kind of overview of how much
energy or co2 depending
on water sometimes also is
being used by the models
for their either training or
their inference or both
mostly training in this case
and in for instance they they

calculate they track monitor
or calculate means that
they actually calculate the
actual numbers they estimate
based on some parameters
they report for instance
if they're only showing a
label like there was in the
previous slide they schedule
they help schedule tasks
for for servers for for hpc
clusters for instance or
they just kind of bring some
labeling to existing data
the end of the energy
consumption the carbon footprint
which is a co2 equivalent
or to both and and this is um
actually very um interesting
because some of them are
using data visualization
some of you might be using
just text and quite a few
of these tools and i'll
discuss some in more depth are
basically a python package
that you can also install
in your in your work that kind
of encapsulates a little bit
your code and then kind
of tracks how how the cpu
your cp local cpu or or remote
if you put the parameters
for remote kind of reacts
to that so one of the
most uh like established
tools like this is called cold
carbon which is meant to
track and reduce your co2
emissions from your computing
and you can see here
in the top it tells you
how much energy is being
computed how many emissions
have been produced and the
cumulative duration this is
this is over time so this
is something that you can
have as a dashboard over
of over your history of your

projects and then this is like an organizational level and it gives you some equivalents to talk to work with like the the houses so how much household weekly per week or how many miles kilometers driven or how many days of television and things like that it also supports this kind of global carbon intensity so if you were indeed going to change your your cloud service then you can kind of choose based on which one has the most green energy at that given moment in time another one is energy viz that is an interactive energy tracking tool for machine learning models it also explains its calculations and specifically it's more it goes more into depth into the united states where it goes state by state and you can actually change the place where you're training or choose alternative hardware or locations based on these kinds of metrics that they're offering.

But it's not just about you training something from Timestamp 17 minutes scratch. There's also this very useful tool called the AI Energy Score that basically helps you decide which foundational models to waste yourself on if you're deciding to fine-tune or if you're deciding to use some kind of rag system where you have text generation image generation classification and other kind of tasks and then it has this leaderboard

explaining how how energy intensive that model has been and this gets updated quite frequently it's from hugging face And I suggest that this is a resource that might be quite useful for you to keep in mind when you're designing projects.

Now, most of the things I've said so far have been about estimations, but quite recently, two companies, Mistral AI and Google Gemini, actually made a peer-reviewed well mistral made a peer-reviewed life cycle assessment for basically going through each stage of the of the life cycle of their of their large uh model mistral large two and then they were calculating to what extent was the the different stages in that model how how much did they affect the overall um greenhouse emissions the water consumptions and the materials and the materials are especially interesting here because these are like the embodied impacts of of using these chips hardwares and so what so on and so forth so as you can see from their life cycle that they have like they start from infrastructure computing and usage then you have most of their emissions come from model training and inference water as well coming from model training and inference and most of the material consumptions happens during the embodied impact So this is about server manufacturing and end of life of those specific servers. And what they found is that

for a typical 400, like
in terms of inference, for
a typical 400 token response
from the Leshya assistant,
Mistral had an impact of
1.14 grams of CO2 equivalent
and around 40 how much
was it 45 milliliters of
water consumed so that's not
it doesn't seem significant
but if you consider how
many people are now embedding
generative AI and inference
in absolutely everything
that is being done and how
many billions of users
there are then these figures
start adding up quite
substantially um yeah the next
very close to each other
like two weeks later google
published their own
measurement of their impact of
their inference stages and
specifically for gemini and
Timestamp 20 minutes they found that they they
have this very interesting
graphs where they they kind
of reverse it a little bit
and they look at how many
prompts per kilowatt hours
are possible to make and
you can see the depending on
the the way that you're
kind of measuring the energy
there's a big variation here
for a median Gemini Gemini
prompt between almost 4
,000 I think it was 4,167
prompts per kilowatt hours all
the way up to 10,000 prompts
per kilowatt hour so
depending on the metrics there
is quite different and
then you can also see some
other tools here like the
average chat gpt which is
coming from some some altman's
blog as an estimate so these
are not measured and

that's why they're in gray
they're just estimates with
different levels of backing
up with from data but i
suggest when you look at these
graphs you always make sure
that they're talking about
measurements rather than
estimates because it becomes
very hard to compare and
then you can see here how
their methodology kind of
divides their prompt into
active accelerator cpu and
an idle time of the of the
servers and whatnot okay so
that's one way of looking
at how um human ai interaction
can support, basically,
environmental sustainability
by looking at ecofeedback
and looking how, like,
people or developers can get
feedback on their actual
energy consumptions, water
consumptions, mineral
consumptions to decide, make
decisions, more environmentally
conscious decisions in
their actual workplace or
individually. So what we also
found, this is something
that is another dimension
that human computer interaction,
human AI interaction
kind of supports quite a lot,
is looking at how perceptions
of AI, how perceptions of
sustainability are when
it comes to AI practitioners.
so basically there's
quite a lot of research already
about how sustainability
is understood within household
contexts but how does
that translate into something
that is more professional
like something that has
to do with your studies
or your work so here i

present it's part of our own research that we've done in kings we've done interviews with machine learning practitioners from academia and industry and look at how the general questions we had for them was about their background their work practices and then looking at how do they include sustainable factors in their work and their perceptions of sustainability policies and industry trends and it was really interesting conversations to have and and it's it's not necessarily relevant for you to know everything here but i just wanted to to focus on on two three interesting points that i think might help you um understand the problem with a little bit more holistically uh for instance like a lot of the people said that they don't do something about sustainability in their work because they don't find it's their responsibility or they don't think that they have a big impact on what they're doing with their choices rather than it's a responsibility of just like bigger companies like open ai the googles the metas and those kinds of scale then quite a lot is also about the uncertainty of calculation measurement and then they talk about different ways that that can can change necessarily for individuals but also for and as a way to demonstrate how research works in hdai i will give you some quotes which is coming again from

interviews that we've done
with these practitioners
and if you remember again on
week five or four then we
were doing this kind of
requirements understanding
of how people react to new
technologies and this is
part of a methodology like
this so some people say that
the effects are maybe very
small so they don't think
that it's something that
is as important as the
importance of machine learning
in general on the other
way around. So rather than
looking at how important,
how impact, how much impact
AI has on the environment,
they think the other way
around works better. That
AI systems actually are
benefiting the environment
more than they're negating it.
Timestamp 25 minutes And then things like a
quote we had like uh if
somebody somebody says oh
why did you use that because
it's environmentally
friendly you know people
are going to laugh at that
they're going to say i
don't care how environmentally
friendly it is i
want to say lives and in
this case it was a person
that was working in the
medical domain so the
priorities seems like for many
people that the importance
of using generative ai
in their lives is much
more crucial to their
this environmental impact.
And then there was
a consistent role of
perception of minimal
role and responsibility
that individuals

have. And this was
kind of discussed in
many different ways.

Like, my work is just
a very small amount in
this unsustainable
system, because there are
people that use much
more powerful models.

Or some of these figures are
a bit shocking, but there
as an individual, you can't
really do much about it
and then they talk about
different types of ways to go
towards change like the
paper argues that individuals
should also have responsibility
at least awareness

of how to continue in this
direction the regulation can
help as a driver and motivator
we need more transparency
on open data and open
data on models because as
I mentioned a lot of these
things are actually estimated
and very few are actually
calculated and the the
recent contributions of Google
and Mistral are actually
very welcome in that
dimension. And then we can also
learn from other topics, like
we talked about responsibility
and fairness in the
previous week. So we can
see how that discourse has
progressed over the years and
see how sustainability can
be part of that as well.

okay so far we looked at
how sustainability relates
to developers and practitioners
which is basically what
a lot of you are training
to become but there's also
another type of users right
when it comes to generative
ai and these are the end

users so like things like
how much does a chat gpt
or similar tool cost in
terms of energy are people
using it aware and do they
mind of this implication of
this kind of usage or this
environmental impact these
are other questions that
human ai interaction can
help kind of address and
here i've been as a this is
a diagram just to mention
that i've been talking
mostly about developers
and their responsibility
now we're going to switch
back to the inference
pipeline and and look at the
usage and end uses so again
here There are some tools
to support this process,
and I kind of really
suggest you get acquainted
with them, because it can
help you figure out and
estimate how impactful your
potential AI systems that
you're designing will be.
This is Ecologites. It's a
calculator, but it's also a
GitHub repository, and
it's also something you can
kind of embed quite easily in
your code. it's not something
that you need to to use
directly on a web app but
I'm using this as an example
to show you how it works
so here what you do is you
have you input the provider
and the model you're talking
about and then how what
kind of prompt are you
expecting to have like a small
conversation with a chat bot
like 400 output tokens and
then it helps you estimate
the energy the greenhouse
emissions the abiotic resources

and the primary energy
that this prompt and this
conversation with that specific
model and type will actually
have so this doesn't
calculate things like image
generation that I mentioned
already is quite more more
energy intensive than than
text generation but still
it's a very good first start
to start including that in
your estimations of your
projects of if you had X
amount of users like you're
hoping to have how much uh
how much energy do you think
or what kind of impact would
that have on the environment
um by incorporating kind
of this kind of model and
here there are different
types of models some are
closed source so they're
all based on estimations but
others are open source so
they're actually there are
actually some measurements
that you can you can work with
their methodology is
also very interested
interesting sorry so
if you're interested in
this topic then i think
it's a it's a it's
a very deep resource
that you can work with
and i like this one also
i mentioned it is not as
complete as ecologies but
i like it this specifically
because it's one of the few
tools that does inference and
it is um it lets you
actually it does measurements
it's not just estimations so
you can use some open source
models like for instance
google's gemma is the one
chosen here and when you

type things here in your prompts and you kind of do your conversation or your queries or whatever then you get this kind of toolbar underneath here as you can see that tells the live total energy the percentage the how much is the percentage of a phone charge and how long did that take to that communication what's the latency basically of the response and the answer or the answer and the response and here it says importantly that the energy consumption is measured directly on the gpu during inference for all the messages so these are not estimations so these are figures that you can actually they're more reliable even though again this is only for open source models this is not for closed source models but it's still a good step forward or a way to consider when you're designing your interactive systems, your AI systems, like how can you incorporate something like this to be more transparent in terms of environmental sustainability in your work.

Okay, and that's the summary of how environmental sustainability kind of relates to generative AI.

There's a lot more topics and I will include some of them as additional readings, but i hope this is a good primer kind of to to get you understanding the the human-centered values another type of human-centered values in the design

of ai systems thank you